

Brewed to Be Random

Modeling Customer Orders at Compose Coffee

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Is Compose Coffee random?

During a week-long break, I observed the number of customer orders per hour at a Compose Coffee location in Seoul. The behavior appears to be random because customers visit the cafe for unpredictable reasons: individual schedules, meetups, spontaneous orders, and so on. The time between each order varies, and there is no coordinated pattern behind who shows up or when.

Describing coffee orders

I recorded the number of orders placed each hour for 7 days from 9 a.m. to 5 p.m., resulting in 56 total hourly time blocks ($7 \text{ days} \times 8 \text{ hour blocks} = 56 \text{ hour blocks}$).

The raw data is shown in Table 1 below. There were 870 orders in total. The distribution of hourly orders is fairly symmetric, as the mean (15.54) and median (15.0) differ by less than half a standard deviation. The variance (12.58) is close to the mean, and because the data consists of independent observations representing the number of events occurring in fixed time intervals, this supports the assumption that the counts follow a Poisson distribution. The data show moderate variability ($SD = 3.55$), a slight right skew (skewness = 0.45), and two potential outliers were detected (23 and 24) using the $1.5 \times IQR$ rule. Hourly order counts range from 9 to 24.

Table 1: Hourly Order Counts at Compose Coffee by Day

Day	9 AM	10 AM	11 PM	12 PM	1 PM	2 PM	3 PM	4 PM
Monday	18	11	20	12	13	15	17	13
Tuesday	15	12	17	17	13	18	10	13
Wednesday	14	16	9	15	15	20	13	11
Thursday	19	19	11	14	21	14	14	19
Friday	22	19	16	16	14	14	21	16
Saturday	12	13	23	20	22	14	12	15
Sunday	10	24	16	15	13	17	12	16

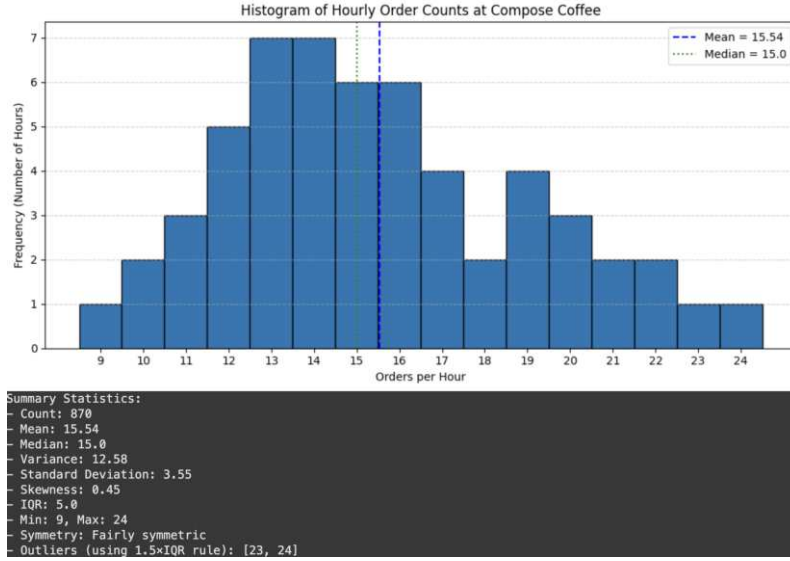


Figure 1: Histogram of Hourly Order Counts at Compose Coffee

Does the data follow a Poisson distribution?

Based on the characteristics of the data, including the similarity of the mean and variance, the fact that I was counting discrete events over fixed intervals, and the assumption that these events occur independently, I hypothesize that the hourly order counts follow a Poisson distribution.

To evaluate this, I perform a chi-squared goodness-of-fit test¹ with a significance level of $\alpha = 0.05$. The test compares observed frequencies of order counts with the expected frequencies under a Poisson model with the same average rate.

The observed mean is $\lambda = 15.54$, which I use as the rate parameter. The hypotheses are:

1. **Null hypothesis** (H_0): The data follows a Poisson distribution with $\lambda = 15.54$.
2. **Alternative hypothesis** (H_A): The data does not follow a Poisson distribution with $\lambda = 15.54$.

Using the Poisson PMF, I calculate expected frequencies for each distinct order count across the 56 hours:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad (1)$$

$$E[X_i] = P(X = k) \cdot n \quad (2)$$

where $n = 56$ is the total number of hours observed. Expected frequencies are normalized to sum to exactly 56 to satisfy the assumptions of the chi-squared test.

¹Shier, R. (2004). Statistics: 1.4 Chi-squared goodness of fit test. *Mathematics Learning Support Centre*. Retrieved from https://www.lboro.ac.uk/media/media/schoolanddepartments/mlsc/downloads/1_4_gofit.pdf

Since the p-value (0.9991) is greater than 0.05, I fail to reject the null hypothesis. There is no statistically significant difference between the observed and expected frequencies, which supports the conclusion that the data follows a Poisson distribution.

Visual check

To support the statistical test, I plot the observed order count frequencies against the Poisson PMF at $\lambda = 15.54$ (see Figure 2). The histogram bars and the PMF curve align closely at the peak, spread, and overall shape, which is consistent with the chi-squared result.

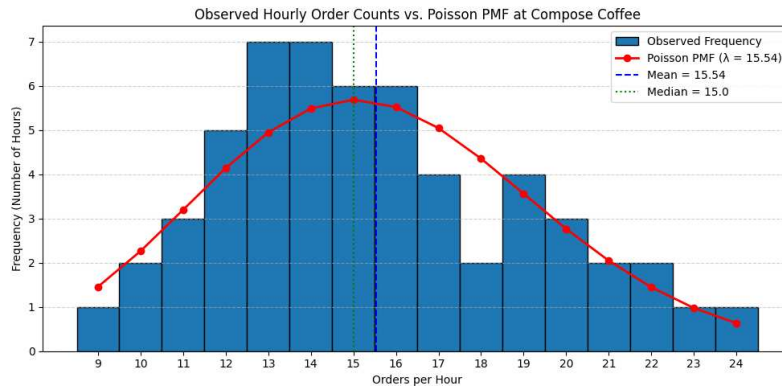


Figure 2: Observed Hourly Order Counts vs. Poisson PMF at Compose Coffee

Summary

Hourly order counts at Compose Coffee closely follow a Poisson distribution. The mean and variance are nearly equal (15.54 and 12.58), the chi-squared test returns a p-value of 0.9991, and the observed distribution visually matches the Poisson PMF. All three lines of evidence point in the same direction.

Appendices

Appendix A: Data loading

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import scipy.stats as stats
from scipy.stats import poisson, chisquare
from collections import Counter

sheet_id = "104SR0o7_dgwgyaK06cv8QzW3FCg1-1F3sPjQITHii38"
url = f"https://docs.google.com/spreadsheets/d/{sheet_id}/export?format=csv"

data = pd.read_csv(url)
```

Appendix B: Descriptive statistics and histogram

```
order_counts = data.drop(columns="Day").values.flatten().tolist()

num_orders = sum(order_counts)
mean_val = np.mean(order_counts)
median_val = np.median(order_counts)
std_dev = np.std(order_counts, ddof=1)
variance_val = np.var(order_counts, ddof=1)
skewness = stats.skew(order_counts)
iqr = stats.iqr(order_counts)
min_val = min(order_counts)
max_val = max(order_counts)

lower_bound = median_val - 1.5 * iqr
upper_bound = median_val + 1.5 * iqr
outliers = [x for x in order_counts if x < lower_bound or x > upper_bound]

plt.figure(figsize=(10, 5))
plt.hist(order_counts,
         bins=range(min_val, max_val + 2),
         edgecolor='black',
         align='left')
plt.xlabel("Orders per Hour")
plt.ylabel("Frequency (Number of Hours)")
plt.title("Histogram of Hourly Order Counts at Compose Coffee")
plt.xticks(range(min_val, max_val + 1))
plt.grid(axis='y', linestyle='--', alpha=0.6)
plt.axvline(mean_val, color='blue', linestyle='dashed', linewidth=1.5,
            label=f"Mean = {mean_val:.2f}")
plt.axvline(median_val, color='green', linestyle='dotted', linewidth=1.5,
            label=f"Median = {median_val}")
plt.legend()
plt.tight_layout()
plt.show()

print(f"Count: {num_orders}")
print(f"Mean: {mean_val:.2f}")
print(f"Median: {median_val}")
```

```

print(f"Variance: {variance_val:.2f}")
print(f"Standard Deviation: {std_dev:.2f}")
print(f"Skewness: {skewness:.2f}")
print(f"IQR: {iqr}")
print(f"Min: {min_val}, Max: {max_val}")
print(f"Outliers: {outliers if outliers else 'None'}")

```

Appendix C: Chi-squared goodness-of-fit test

```

n = len(order_counts)
lam = np.mean(order_counts)

count_range = range(min(order_counts), max(order_counts) + 1)
observed_counts = Counter(order_counts)
observed = np.array([observed_counts.get(k, 0) for k in count_range])

expected_probs = poisson.pmf(count_range, mu=lam)
expected = expected_probs * n
expected = expected * (n / expected.sum())

chi2_stat, p_value = chisquare(f_obs=observed, f_exp=expected)
print("Chi-squared statistic:", round(chi2_stat, 2))
print("P-value:", round(p_value, 4))

```

Chi-squared statistic: 3.41

P-value: 0.9991

Appendix D: Observed vs. Poisson PMF plot

```

plt.figure(figsize=(10, 5))
plt.hist(order_counts,
         bins=range(min_val, max_val + 2),
         edgecolor='black',
         align='left',
         label='Observed Frequency')

x_vals = np.arange(min_val, max_val + 1)
poisson_probs = poisson.pmf(x_vals, mu=mean_val)
expected_freqs = poisson_probs * len(order_counts)
plt.plot(x_vals, expected_freqs, 'o-', color='red', linewidth=2,
         label=f'Poisson PMF (lambda = {mean_val:.2f})')

plt.axvline(mean_val, color='blue', linestyle='dashed', linewidth=1.5,
            label=f"Mean = {mean_val:.2f}")
plt.axvline(median_val, color='green', linestyle='dotted', linewidth=1.5,
            label=f"Median = {median_val}")

plt.xlabel("Orders per Hour")
plt.ylabel("Frequency (Number of Hours)")
plt.title("Observed Hourly Order Counts vs. Poisson PMF at Compose Coffee")
plt.xticks(x_vals)
plt.grid(axis='y', linestyle='--', alpha=0.6)
plt.legend()
plt.tight_layout()

```

```
plt.show()
```
